NEWSQL
The New Way to Handle Big Data

Big data, big data, big data! This term has been dominating information management for a while, leading to enhancements in systems, primarily databases, to handle this revolution. Though there are many alternative information management systems available for users, in this article, the authors share their perspective on a new type, termed NewSQL, which caters to the growing data in OLTP systems.

Everywhere we look in today's data management landscape, the volume of information is soaring. According to one estimate, the data created in 2010 is about 1200 exabytes, and will grow to nearly 8000 exabytes by 2015, with the Internet/ Web being the primary data driver and consumer. This growth is outpacing the growth of storage capacity, leading to the emergence of information management systems where data is stored in a distributed way, but accessed and analysed as if it resides on a single machine (Figure 1).

Besides resolving the data size problem, these solutions also need to cater to massive performance requirements to ensure timeliness of data processing. Unfortunately, an increase in processing power does not directly translate to faster data access, triggering a rethink on existing database systems. To understand the enormity of data volumes, let's look at a couple of figures: Facebook needs to store 135 billion messages a month. Twitter has the problem of storing 7 TB of data per day, with the prospect of this requirement doubling multiple times per year. Criticality of data and continuity in data availability has become more important than ever. We expect data to be available 24x7 and from everywhere. This brings in the third dimension of high availability and durability. So we have to factor in high availability without any single point of failure, which has traditionally been a telecom forte. These challenges have created a new wave in database processing solutions, which manage data in both structured and unstructured ways.

Legacy information management systems are characterised by monolithic relational databases, disk/tape-based stores (as memory in huge quantities is scarce or expensive), vendor lock-in and limited enterprise-level scalability. With the advent of the cloud, new data management solutions are emerging to handle distributed (relational/non-relational) content on open platforms at the speed of a mouse-click.

NoSQL

One of the key advances in resolving the 'big-data' problem has been the emergence of NoSQL as an alternative database technology. NoSQL (sometimes expanded to "not only SQL") is a broad class of DBMS that differ significantly from the classic RDBMS model. These data stores may not require fixed table schemas, usually avoid *join* operations and typically scale horizontally.

NoSQL solutions have carved a niche market for themselves as key-value stores, document databases, graph databases and big-tables. Use cases for these solutions in Web applications are a dime a dozen. They are used everywhere—whether it is in Web-based social interactions like online gaming/networking, or revenue maximising decisions like ad offerings, and not to mention basic operations like searching the Internet. Figure 2 shows the architecture of a famous NoSQL database (CouchDB) written in Erlang.

But, is NoSQL the answer to all problems? In spite of the technological acumen provided by NoSQL solutions, the RDBMS users in enterprises are reluctant to switch to it. Why?